

7. Project Documentation

FitBuddy — User guide, setup guide, API reference, and operational notes

Project Overview

FitBuddy is a FastAPI + Jinja2 + SQLite application that generates personalized seven-day workout plans and nutrition/recovery tips, then allows users to refine the workout plan through feedback.

Installation — Windows

```
py -m venv venv
venv\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt
copy .env.example .env
```

Add the Gemini API key to .env when live AI generation is required.

Running the Application

```
uvicorn app.main:app --reload
```

- Home: <http://127.0.0.1:8000>
- Swagger/API docs: <http://127.0.0.1:8000/docs>
- Admin/coach dashboard: <http://127.0.0.1:8000/view-all-users>
- Health: <http://127.0.0.1:8000/health>

Using the Web Interface

- Enter name and unique user ID.
- Enter age and weight in kilograms.
- Select fitness goal, workout intensity, and experience level.
- Choose Generate My Plan.
- Review the seven-day plan and nutrition/recovery tip.
- Enter feedback when changes are required and submit the update.

API Examples

```
POST /api/generate-workout
{
  "user_id": "FB001",
  "username": "Alex",
  "age": 21,
  "weight": 65,
  "goal": "muscle gain",
  "intensity": "medium",
  "experience": "beginner"
}
```

```
POST /api/submit-feedback
{
  "user_id": "FB001",
  "feedback": "Add more cardio and keep Day 4 as a full rest day."
}
```

Operational Notes

- Set DATABASE_URL to change the database location; the default is SQLite at ./fitbuddy.db.
- Set GOOGLE_API_KEY for live Gemini use.
- Keep API keys out of source control.
- The README explicitly positions FitBuddy as wellness guidance rather than a medical diagnostic or treatment system.

Project basis: This document is derived from the supplied FitBuddy source package, including its README, FastAPI routes, SQLAlchemy models, Jinja2 templates, CSS, and tests.